Manifold-Informed Architecture Benchmark

Dataset: CIFAR100 | Generated: 2026-03-29 00:35:26

Executive Summary

```
Dataset           :  CIFAR100
Input dimension   :  3,072D  (ambient pixel space)
Classes           :  100
Intrinsic dim (d) :  34  (98.9% of ambient dims are noise)
Variance threshold:  τ = 0.9
```

Training Protocol

```
Epochs            :  50
Trials            :  3
Optimizer         :  Adam
```

Best Architecture

```
Intrinsic Dim (PCA→100D→output)
Accuracy  :  0.2540 ± 0.0010
Params    :  20,200
Eff       :  0.0126 acc / Kparam
```

vs Standard Architecture

```
Accuracy delta    :  +0.0648  (+6.48 pp)
Parameter ratio   :  85.7x fewer parameters
Efficiency gain   :  115.0x acc/Kparam
```

Provenance & Environment

Repository

```
Project  :  proteusPy
URL      :  https://github.com/suchanek/proteusPy
Commit   :  d75f66ee  (--abbrev-re
d75f66ee23710c2532ea5fe46bd3588c95e40517)
Date     :  2026-03-29 00:28:46 -0400
Message  :  add: unified test structure, outputs
```

Software

```
Python       :  3.12.13
TensorFlow   :  2.16.2
Platform     :  macOS-26.4-arm64-arm-64bit
Host         :  Turing
```

Hardware

```
Machine      :  Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Compute      :  CPU
```

Manifold Discovery Parameters

```
k neighbors (local PCA)  :  ?
Discovery samples        :  ?
Samples per class        :  ?
Variance threshold τ     :  0.9
```

Results Files

```
JSON   :  cifar100_architecture_results.json
Figure :  cifar100_architecture_results.png
Report :  cifar100_report.pdf
Report :  cifar100_report.md
```

```
Report generated  :  2026-03-29 00:35:26
```

Manifold Discovery

Local PCA | k=? | τ=0.9

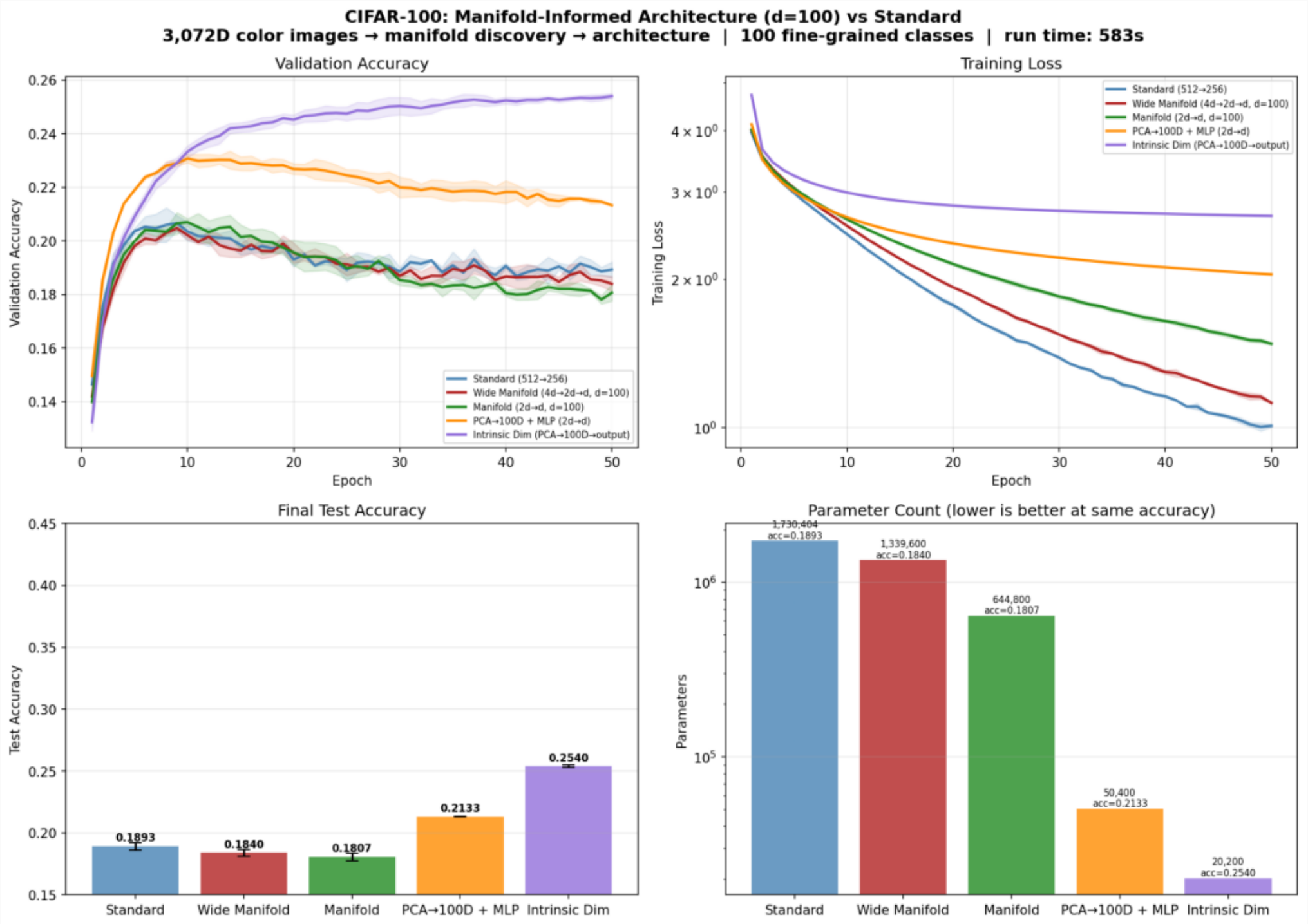
Intrinsic Dimensionality vs Variance Threshold

```
τ    Mean d     Std    Min    Max   Noise %
-----------------------------------------------
0.95      34.5     2.3     26     38     98.9%
0.90      27.1     2.4     19     31     99.1%
0.85      22.0     2.3     14     26     99.3%
0.80      18.1     2.2     11     22     99.4%

Selected d = 34  (max per-class max at τ=0.9, clamped to n_classes=100)
```

Per-Class Intrinsic Dimensionality

```
(Top 10 hardest + 10 easiest of 100 classes)

Class   Mean d    Std   Min   Max
-----------------------------------------------
tiger     32.3    0.6    31    33
snake     32.2    0.6    31    33
motorcycle     32.1    1.0    30    33
streetcar     31.7    0.8    30    33
mushroom     31.6    1.4    29    33
butterfly     31.2    2.8    26    34
bus     31.1    0.9    29    32
tractor     31.0    1.0    30    32
raccoon     30.8    1.9    28    33
fox     30.7    1.0    29    32
...      ...
cockroach     22.3    1.7    19    25
cup     22.3    1.1    20    24
plate     22.1    1.1    20    24
shark     21.6    0.7    20    22
ray     21.4    1.5    19    24
bottle     21.1    1.6    19    23
rocket     21.0    1.3    19    23
cloud     19.5    1.0    18    21
plain     17.2    1.7    13    19
sea     16.3    0.9    15    18
```

CIFAR100 — Architecture Comparison Results



Architecture Comparison

CIFAR100 | 50 epochs | 3 trials | Adam optimizer

Results Table

Architecture	Params	Test Acc (mean±std)	Loss	Acc/Kparam
Standard (512→256)	1,730,404	0.1893 ± 0.0028	9.0235	0.0001
Wide Manifold (4d→2d→d, d=100)	1,339,600	0.1840 ± 0.0028	7.7288	0.0001
Manifold (2d→d, d=100)	644,800	0.1807 ± 0.0031	6.7401	0.0003
PCA→100D + MLP (2d→d)	50,400	0.2133 ± 0.0000	4.2884	0.0042
✦Intrinsic Dim (PCA→100D→output)	20,200	0.2540 ± 0.0010	3.2362	0.0126

✦ = best performing architecture

Parameter Efficiency Ranking (acc / Kparam)

1. Intrinsic Dim (PCA→100D→output)	0.0126	(0.2540 / 20,200)
2. PCA→100D + MLP (2d→d)	0.0042	(0.2133 / 50,400)
3. Manifold (2d→d, d=100)	0.0003	(0.1807 / 644,800)
4. Wide Manifold (4d→2d→d, d=100)	0.0001	(0.1840 / 1,339,600)
5. Standard (512→256)	0.0001	(0.1893 / 1,730,404)

Summary vs Standard

Best architecture	:	Intrinsic Dim (PCA→100D→output)
Accuracy delta	:	+0.0648 (+6.48 pp)
Parameter ratio	:	85.7x fewer parameters
Efficiency gain	:	115.0x acc / Kparam

Key Findings & Interpretation

Manifold Hypothesis Test Results

The CIFAR100 training set occupies a 34-dimensional manifold
embedded in 3,072-dimensional pixel space.
98.9% of ambient dimensions carry no structure.

Architecture Performance

A manifold-informed architecture (Intrinsic Dim (PCA→100D→output))
achieves 0.2540 test accuracy — 6.48 pp better
than the Standard baseline (0.1893),
using 85.7x fewer parameters (20,200 vs 1,730,404).

Parameter efficiency improves 115.0x over Standard.

PCA Pre-Projection Insight

When ambient noise exceeds 90%, PCA pre-projection to d=34
removes uninformative variance before the network trains.
Networks operating on raw 3,072D input must learn this
projection implicitly, wasting capacity on noise dimensions.

Interpretation

These results support the manifold hypothesis for classification:
sizing network bottlenecks to the intrinsic dimensionality of the
data manifold — rather than the ambient input dimension — yields
equal or better accuracy with substantially fewer parameters.

The parameter efficiency advantage scales with the ambient-to-
intrinsic dimensionality ratio (compression factor).